

LAB NOC 06

Cisco Catalyst 2960 — MAC Address Tracking

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Ticket ID	NOC-006	NOC-006
Categoría / Category	Seguridad — Rastreo de dispositivos por MAC	Security — Device tracking by MAC address
Nivel / Level	CCNA — Switching / Seguridad L2	CCNA — Switching / L2 Security
Duración / Duration	Aprox. 20 minutos	Approx. 20 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Relevancia NOC	Identificación de dispositivos no autorizados, forense L2	Unauthorized device identification, L2 forensics

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Localizar un dispositivo desconocido en la red usando únicamente su MAC address. Ejecutar show mac address-table para identificar el puerto del switch donde está conectado, confirmar el estado físico del puerto, y documentar el hallazgo en formato de Incident Report NOC.	Locate an unknown device on the network using only its MAC address. Execute show mac address-table to identify the switch port where it is connected, confirm physical port status, and document the finding in NOC Incident Report format.

INCIDENT TICKET — NOC-006

CAMPO / FIELD	DETALLE / DETAIL
Incident ID	NOC-006
Title / Título	Unknown device connected to network / Dispositivo desconocido conectado a la red
Priority / Prioridad	Medium — posible dispositivo no autorizado / possible unauthorized device
Service / Servicio	Network Security — Layer 2
Device / Equipo	Cisco Catalyst 2960 — Access Layer Switch
Description	Dispositivo detectado en la red. Se necesita identificar el puerto del switch donde está conectado usando solo la MAC address. / A device has been detected on the network. Need to identify the switch port

	where it is connected using only the MAC address.
MAC Address	c4da.2604.8923
Investigation Method	show mac address-table → localizar MAC → confirmar puerto con show interfaces status
Finding	Dispositivo ubicado en puerto Fa0/1 — estado: connected / Device located on port Fa0/1 — status: connected
Status	CERRADO — dispositivo localizado y documentado / CLOSED — device located and documented

PROCEDIMIENTO — 4 ETAPAS / PROCEDURE — 4 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Recibir ticket NOC-006: dispositivo desconocido detectado en la red con MAC c4da.2604.8923	STAGE 1 — Receive ticket NOC-006: unknown device detected on network with MAC c4da.2604.8923
2	ETAPA 2 — Ejecutar: show mac address-table en el Cisco Catalyst 2960	STAGE 2 — Run: show mac address-table on the Cisco Catalyst 2960
3	Localizar la entrada: VLAN 1 c4da.2604.8923 DYNAMIC Fa0/1	Locate entry: VLAN 1 c4da.2604.8923 DYNAMIC Fa0/1
4	Identificar: dispositivo conectado en puerto Fa0/1, VLAN 1, aprendizaje dinámico	Identify: device connected on port Fa0/1, VLAN 1, dynamic learning
5	ETAPA 3 — Confirmar estado físico: show interfaces status include Fa0/1	STAGE 3 — Confirm physical status: show interfaces status include Fa0/1
6	Resultado: Fa0/1 connected — dispositivo físicamente conectado y activo	Result: Fa0/1 connected — device physically connected and active
7	ETAPA 4 — Documentar hallazgo en ticket NOC-006 y cerrar con evidencia	STAGE 4 — Document finding in ticket NOC-006 and close with evidence

LAB 06 — MAC Address Tracking

Equipo: Cisco Catalyst 2960

Ticket NOC

Incident ID: NOC-006**Title:** Unknown device connected to network**Description**

A device has been detected on the network.

We need to identify the switch port where it is connected.

Objetivo Encontrar: qué puerto del switch tiene conectado el dispositivo usando solo: MAC address. Command: show mac address-table

```
COM5 - PuTTY
Switch>en
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
A11     0100.0ccc.cccc   STATIC  CPU
A11     0100.0ccc.cccd   STATIC  CPU
A11     0180.c200.0000   STATIC  CPU
A11     0180.c200.0001   STATIC  CPU
A11     0180.c200.0002   STATIC  CPU
A11     0180.c200.0003   STATIC  CPU
A11     0180.c200.0004   STATIC  CPU
A11     0180.c200.0005   STATIC  CPU
A11     0180.c200.0006   STATIC  CPU
A11     0180.c200.0007   STATIC  CPU
A11     0180.c200.0008   STATIC  CPU
A11     0180.c200.0009   STATIC  CPU
A11     0180.c200.000a   STATIC  CPU
A11     0180.c200.000b   STATIC  CPU
A11     0180.c200.000c   STATIC  CPU
A11     0180.c200.000d   STATIC  CPU
A11     0180.c200.000e   STATIC  CPU
A11     0180.c200.000f   STATIC  CPU
--More--
```

```
Switch#
Switch#
Switch#show mac address-table | include Fa0
  1    c4da.2604.8923    STATIC    Fa0/1
Switch#
```



Interpretación NOC

1) Encuentro la MAC

MAC = c4da.2604.8923

2) Veo el puerto

PORT = Fa0/1

Conclusión: el dispositivo está conectado en el puerto Fa0/1

Conclusion: The device is connected and generating traffic on port Fa0/1.

3) Confirmación

El NOC normalmente revisa el puerto:

show interfaces status | include Fa0/1

Resultado típico:

Fa0/1 connected

Entonces confirma:

Device located on port Fa0/1

Respuesta final al ticket

El NOC documenta algo así:

Incident

Unknown device detected in network
MAC: C4DA.2604.8923

Investigation

show mac address-table | include Fa0

Result

Device located on switch port Fa0/1

Se usa para:

- encontrar PCs
- encontrar cámaras
- encontrar access points
- detectar dispositivos no autorizados

Esto es **MAC tracking**, una tarea muy común en NOC.

LOGICA NOC — 3 PASOS DE MAC TRACKING / NOC LOGIC — 3 MAC TRACKING STEPS

#	PASO (ES)	ACCION (ES)	STEP (EN)	ACTION (EN)
1	Encontrar la MAC en la tabla	Ejecutar show mac address-table — localizar la entrada con la MAC objetivo: c4da.2604.8923	Find the MAC in the table	Run show mac address-table — locate the entry with target MAC: c4da.2604.8923
2	Identificar el puerto	La tabla muestra: VLAN 1, MAC c4da.2604.8923, tipo DYNAMIC, puerto Fa0/1 → dispositivo conectado en Fa0/1	Identify the port	Table shows: VLAN 1, MAC c4da.2604.8923, type DYNAMIC, port Fa0/1 → device connected on Fa0/1

3	Confirmar estado físico	show interfaces status include Fa0/1 → resultado: Fa0/1 connected → confirmado	Confirm physical status	show interfaces status include Fa0/1 → result: Fa0/1 connected → confirmed

CAMPOS DE show mac address-table / show mac address-table FIELDS

CAMPO / FIELD	SIGNIFICADO (ES)	MEANING (EN)
VLAN	VLAN donde se aprendió la MAC — útil para determinar segmento de red	VLAN where MAC was learned — useful for determining network segment
Mac Address	Dirección MAC del dispositivo (formato xxxx.xxxx.xxxx en Cisco)	Device MAC address (Cisco format xxxx.xxxx.xxxx)
Type	DYNAMIC = aprendida automáticamente / STATIC = configurada manualmente (Port Security sticky)	DYNAMIC = learned automatically / STATIC = manually configured (Port Security sticky)
Ports	Puerto físico del switch donde está conectado el dispositivo	Physical switch port where the device is connected

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>show mac address-table</code>	Muestra toda la tabla MAC dinámica del switch	Shows full dynamic MAC table of the switch
<code>show mac address-table include <MAC></code>	Filtra la tabla por MAC específica — búsqueda directa	Filters table by specific MAC — direct search
<code>show interfaces status include Fa0/1</code>	Confirma estado físico del puerto encontrado	Confirms physical status of the found port
<code>show mac address-table include Fa0/1</code>	Muestra todas las MACs aprendidas en un puerto específico	Shows all MACs learned on a specific port
<code>show mac address-table count</code>	Muestra total de entradas MAC por VLAN	Shows total MAC entries per VLAN
<code>clear mac address-table dynamic</code>	Limpia la tabla MAC — los dispositivos activos re-aprenden inmediatamente	Clears MAC table — active devices re-learn immediately

CASOS DE USO EN NOC / NOC USE CASES

TIPO DE DISPOSITIVO / DEVICE TYPE	CASO DE USO (ES)	USE CASE (EN)
PC / Workstation	Usuario reporta sin conectividad — rastrear por MAC para ubicar puerto físico	User reports no connectivity — track by MAC to locate physical port

IP Camera	Camara IP desaparece del NVR — verificar si el puerto del switch sigue activo	IP camera disappears from NVR — verify if switch port is still active
Access Point	AP reportado offline — confirmar si sigue conectado al switch	AP reported offline — confirm if still connected to switch
Dispositivo no autorizado	Alerta de seguridad — identificar puerto exacto para desconexión inmediata	Security alert — identify exact port for immediate disconnection
PLC / SCADA device	Dispositivo OT sin respuesta — verificar conectividad L2 antes de escalar a OT team	OT device not responding — verify L2 connectivity before escalating to OT team

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
La MAC aparece como DYNAMIC — significa que el switch la aprendió automáticamente cuando el dispositivo envió tráfico. No fue configurada manualmente. / MAC appears as DYNAMIC — switch learned it automatically when device sent traffic. Not manually configured.	Información válida para el ticket. DYNAMIC confirma que el dispositivo está activo y comunicándose en la red. Si apareciera como STATIC indicaría Port Security sticky del LAB 02. / Valid ticket information. DYNAMIC confirms device is active and communicating. If STATIC it would indicate Port Security sticky from LAB 02.
Si la MAC no aparece en la tabla, el dispositivo puede estar apagado, en otra VLAN, o conectado a otro switch upstream. / If MAC is not in the table, device may be powered off, on another VLAN, or connected to another upstream switch.	Verificar con show mac address-table en otros switches del path. Si el dispositivo está en otra VLAN, verificar routing. Si está apagado, el ticket requiere intervención física. / Verify with show mac address-table on other switches in the path. If on another VLAN, check routing. If powered off, ticket requires physical intervention.
El mismo puerto Fa0/1 puede aparecer con múltiples MACs si hay un hub o switch no gestionado conectado. / The same port Fa0/1 may show multiple MACs if an unmanaged hub or switch is connected.	Múltiples MACs en un solo puerto es una señal de alerta en NOC — puede indicar un switch no autorizado o hub que burla el Port Security del LAB 02. Escalar para inspección física. / Multiple MACs on a single port is a NOC alert — may indicate an unauthorized switch or hub bypassing Port Security from LAB 02. Escalate for physical inspection.

RESULTADO / RESULT

RESULTADO (ES) Dispositivo desconocido localizado mediante show mac address-table. MAC c4da.2604.8923 identificada en puerto Fa0/1, VLAN 1, estado connected. Ticket NOC-006 cerrado con evidencia documentada.	RESULT (EN) Unknown device located using show mac address-table. MAC c4da.2604.8923 identified on port Fa0/1, VLAN 1, status connected. Ticket NOC-006 closed with documented evidence.
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KEY TAKEAWAY

"MAC tracking es la tecnica de investigacion L2 mas usada en NOC. Con solo dos comandos (show mac address-table + show interfaces status) se puede ubicar cualquier dispositivo en la red en menos de un minuto. En seguridad, permite localizar dispositivos no autorizados para desconectarlos antes de que comprometan la red. En mineria, permite identificar PLCs, camaras y APs que dejan de responder sin necesidad de inspeccion fisica inmediata."

"MAC tracking is the most-used L2 investigation technique in NOC. With just two commands (show mac address-table + show interfaces status) any device on the network can be located in under a minute. In security, it allows locating unauthorized devices for disconnection before they compromise the network. In mining, it allows identifying PLCs, cameras and APs that stop responding without requiring immediate physical inspection."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), the network supports both IT devices (PCs, phones) and OT devices (PLCs, SCADA terminals, gas sensors, IP cameras, emergency radios). MAC address tracking is the first-line technique for: (1) locating a PLC that stopped responding to SCADA — verify it is still connected at L2 before involving the OT team, (2) identifying unauthorized devices plugged into network drops in restricted areas (underground tunnels, processing plants), (3) confirming that IP cameras are physically connected before escalating to the NVR team, and (4) building a physical inventory of all devices per switch port. Combined with Port Security (LAB 02), MAC tracking is the foundation of L2 network security in industrial environments.